



Welcome to class!

In today's class, we will be talking about the digitalization of data. Enjoy the class!

Digitalization of Data

Digitalization is the process of converting information into digital format. This information may represent an object, image, sound, document or a signal (usually an analog signal) organized into a discrete set of its points or samples. This is the binary data that computers and many devices with computing capacity (such as digital camera and digital hearing aids) can process.

Digitalization can also be defined as the integration of digital technologies into everyday life. Digital system uses a binary numeric system in which electronic pulses are represented by either 0 for a Low pulse or 1 for a High pulse. Digital can more easily represent symbols such as alphanumeric characters that represent real-world data than the analog system.

Benefits of digitalization

1. Long term preservation of documents
2. Orderly archiving of documents
3. Easy and customized access to information
4. Easy information dissemination through images and text, CD-ROM, Internet, Intranet and extranets.

Types of digital computer

1. Microcomputers
2. Minicomputers
3. Mainframe computers
4. Supercomputers

The technology of different information age

The ages are:

1. Stone age
2. Iron age
3. Middle age
4. Industrial age
5. Electronic age



Information age	Tools used	Purpose	Period	Examples of tools in that age
Stone age	Stone	Sewing, cutting, counting, defence, transaction, storage, pottery exhibitions.	Below 12 th century	Basalt, sandstone flint etc.
Iron age	Iron	Defence, Agric	12 th century	Hoes and cutlass
Middle age	Writing materials	Knowledge transfer, education	12 th and 13 th century	Pen feather etc
Industrial age	Coals	Power development, faster movement	Late 18 th and early 19 th century	Cars, Ships etc
Electronic age	Computer	Storage, accuracy, speed. Timeliness	Late 19 th century and above	Circuit, Processor.

Evaluation

1. State the different information ages.
2. State the tools used in each age.

Early counting devices

1. Fingers and Toes
2. Stone
3. Sticks
4. Pebbles
5. Cowries

In the early days of man's existence on earth, counting and simple arithmetic were performed using different parts of the body and some other counting aids. The following are devices used by people of ancient times.

Fingers and toes:

In ancient times, calculations were done by the use of fingers and toes. Fingers were used for simple addition and toes were used together with fingers to count up to twenty. The following problems were faced using fingers and toes to count:



1. They could not be used conveniently to count numbers more than twenty.
2. The result obtained from counting with fingers could not be easily remembered.

Stones and pebbles:

Stones were introduced for counting because of the limitations of counting with fingers and toes. Counting with stones involves building a pile of stones where each represents a quantity. The following were problems faced using stones as counting devices:

1. Large numbers of stones were too heavy to carry from one place to another.
2. It was cumbersome to count if the counting process involved large numbers.

Grains:

The use of grains was introduced because of the heaviness of stones. Grains can be kernel, beans, rice, corn etc. Grains were used the same way as stones.

Sticks:

These are small pieces of wood that are used for counting in place of stones and grains. Children in nursery and primary schools use match sticks for counting.

Marks on the wall

This involves the use of sharp objects in drawing a line on the wall for counting.

Continuous marks on walls will make a wall dirty.

Disadvantages of early counting devices

1. They are difficult to carry about.
2. Counting and calculation take a lot of time.
3. They are prone to mistakes.
4. They cannot be used to count or calculate large numbers.
5. Their results cannot be easily remembered.
6. They have no storage facilities.

General evaluation

1. Mention types of computer according to sizes.
2. Which of the size is used for the weather forecast?
3. Another name for mainframe computer is
4. List any two examples of minicomputers.
5. Mention the four categories of computers.



A Handbook on computer studies for schools, pgs 1-3, by Niji Adekolegan.

Weekend assignment

1. The ancient man began counting by using A. fingers B. slide rule C. abacus D. none
2. The following are examples of early counting methods except A. calculator B. pebbles C. grains D. none
3. In using fingers and toes, you can only count up to A. 20 B. 40 C. 50 D. 100
4. The following can be used for counting in the ancient times except for..... A. fingers B. computer C. stones D. none
5. The stages of development of the computer machine are known as..... A. computer age B. computer history C. computer generation D. all

Theory

1. Mention any THREE early counting devices.
2. List any TWO disadvantages of early counting devices.

In our next class, we will be talking more about the **Digitalization of Data**. We hope you enjoyed the class.

